

Project Executable Files

Date	2 July 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	3 Marks

Project Executable Files

This section requires submission of all executable and deployable files for your project. Ensure all files are properly organized, named, and uploaded to the specified submission platform. The evaluator must be able to run or deploy your solution using the provided files and instructions.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes

4	Database schema / seed files (if applicable)	NA
5	Environment configuration file (.env.example similar) or	NA
6	Deployed application URL (if hosted)	No
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA
8	Dockerfile / Containerization config (if applicable)	NA
9	Test files and test results	Yes
10	Demo video or walkthrough (if required)	Yes

Step 2: File / Folder Structure

Provide an overview of your project's file and folder structure below:

```

OptiCrop/
|
└─ app.py

```

```
└─ model/
  -> Label_encoder
  -> metrics.csv
  -> model.pkl
  -> scaler.pkl
-> notebooks/
  ➔ Crop Analysis.ipynb

└─ Crop_recommendation.csv
└─ requirements.txt
└─ README.md

└─ templates/

  ➔ index.html
  about.html
findyourcrop.html
|
└─ static/
|   └─ css/
|   └─ images/
|   └─ js/
|
└─ notebooks/
|   └─ Crop_Analysis.ipynb
|
└─ screenshots/
|   └─ HomePage.png
|   └─ AboutPage.png
|   └─ PredictionResult.png
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	Hosted Link : https://opticrop-ai.onrender.com/
Login Credentials (Demo)	UserName: Jaya12@gamil.com Password: Jaya123
Platform / Hosting Provider	Render
Repository Link	https://github.com/kollaaditya/OptiCrop-AI
Demo Video Link	https://drive.google.com/file/d/1eAUp9IliGkiE8Q3YdEzdXGRs-yOJgM4e/view?usp=sharing

Step 4: Run Instructions

Provide clear, step-by-step instructions for the evaluator to run your project locally:

Pre-requisites

- Python 3.12 or above
- Anaconda Navigator
- Visual Studio Code or PyCharm
- Flask
- Pandas

- NumPy
- Scikit-learn
- Matplotlib
- Seaborn

Step 1: Clone the repository from GitHub.

Step 2: Open the project folder in Visual Studio

Code. Step 3: Install dependencies: `pip install -r requirements.txt`

Step 4: Ensure model.pkl and Crop_recommendation.csv are present.

Step 5: Run the application:

`python app.py`

Step 6: Open browser and visit:

`http://127.0.0.1:5000/`

Step 7: Enter agricultural parameters and click Predict to get crop recommendations.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Application works only on local server	I deployed later on Render
2	Dataset size is limited	Larger datasets can be added in future

3	No user authentication module	Can be implemented in future versions
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